Fatima Jinnah Women University
Opening Portals of Excellence Through Higher Education

# *Assignment 01*

Git, Gitea, GitHub, Git LFS, and GitHub Pages

Solution document

| | |
|---|---|
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| **Registration number** | FA24B1-SE-001 |
| **Section** | A |
| **Course** | Cloud Computing (BSE-410) |
| **GitHub username** | ayeshaaafifa |
| **Ubuntu prompt** | ayeshaaafifa@ubuntu:~$ |
| **Deadline** | 10 October 2026 |

## Submission links

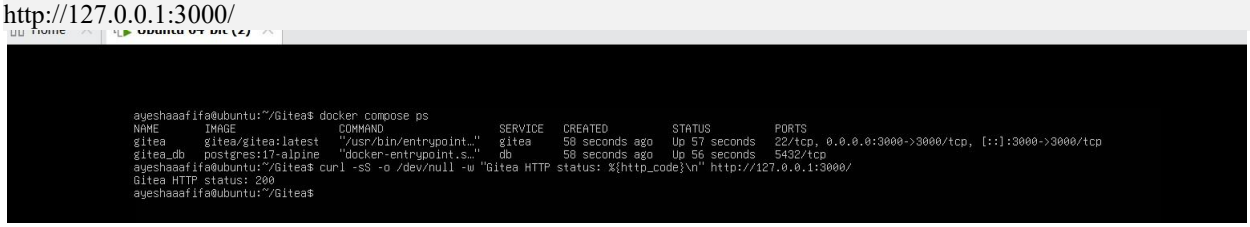
| | |
|---|---|
| **CC repository** | https://github.com/ayeshaaafifa/CC |
| **GitHub Assignment01** | https://github.com/ayeshaaafifa/Assignment01 |
| **Portfolio (GitHub Pages)** | https://ayeshaaafifa.github.io/ |

# Task 1: Install Gitea and Push a Repository from the Ubuntu Server

## Step 1.1: Install and Run Gitea on the Ubuntu Server

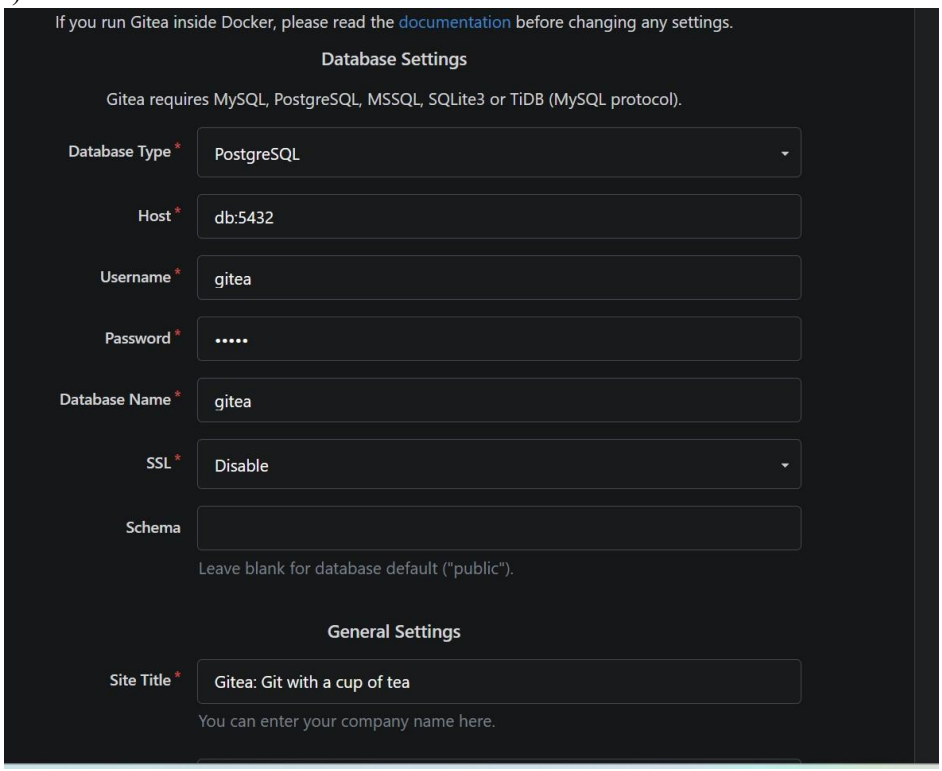
Installed Docker and the Compose plug-in, cloned the instructor's Gitea repository, started Gitea and PostgreSQL, and confirmed Gitea answers on port 3000.

```
git clone https://github.com/WaqasSaleem97/Gitea.git
docker compose up -d docker
compose ps
curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n"
http://127.0.0.1:3000/
```



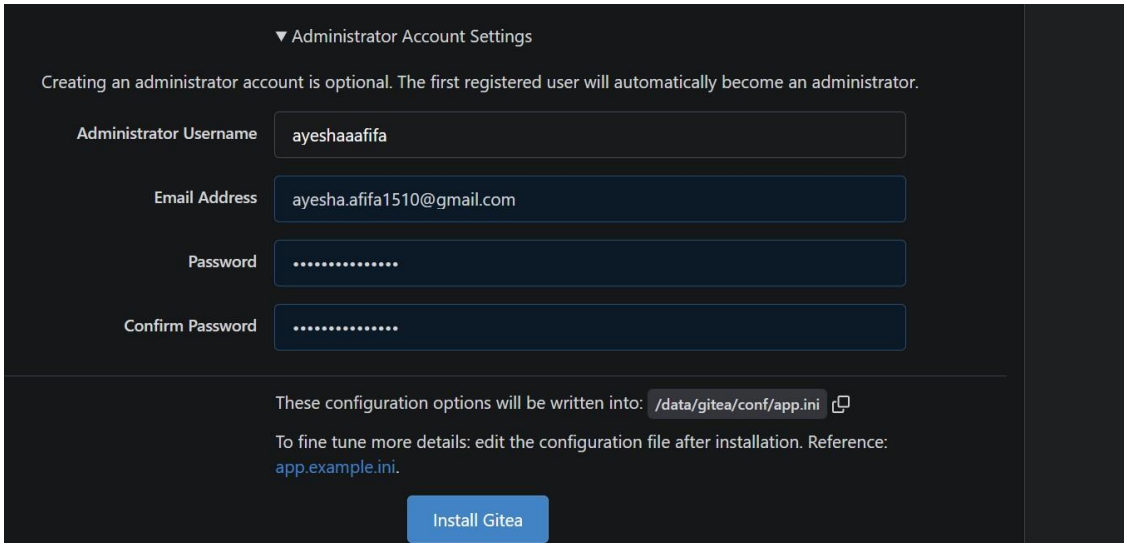

*task1_gitea_running.png: both containers running and Gitea HTTP status 200*

Gitea installation page, using the required PostgreSQL database settings (host db:5432, user gitea, database gitea).




*Gitea installation page: database settings*

Administrator account created to complete the installation.




*Gitea installation page: administrator account*

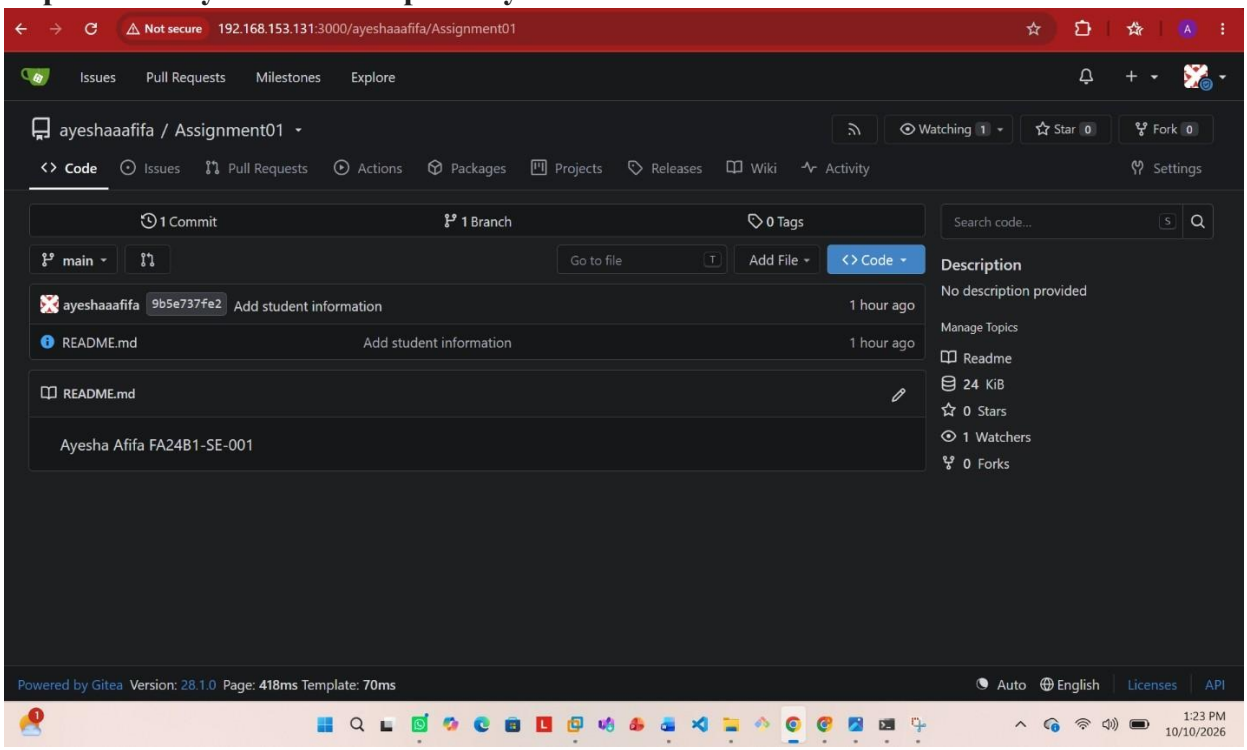
## Step 1.2: Create and Push the Initial Repository

Created an empty public Gitea repository named Assignment01. On the server, created a separate local repository with a README containing my name and registration number, added the gitea remote without a token in the URL, and pushed with a personal access token entered at the password prompt.

```
ayeshaaafifa@ubuntu:~/Assignment01$ git config --global user.name "ayeshaaafifa"
ayeshaaafifa@ubuntu:~/Assignment01$ git config --global user.email "ayesha.afifa1510@gmail.com"
ayeshaaafifa@ubuntu:~/Assignment01$ git commit -m "Add student information"
[main (root-commit) 9b5e737] Add student information
 1 file changed, 2 insertions(+)
 create mode 100644 README.md
ayeshaaafifa@ubuntu:~/Assignment01$ git remote add gitea http://ayeshaaafifa@127.0.0.1:3000/ayeshaaafifa/Assignment01.git
ayeshaaafifa@ubuntu:~/Assignment01$ git remote -v
gitea   http://ayeshaaafifa@127.0.0.1:3000/ayeshaaafifa/Assignment01.git (fetch)
gitea   http://ayeshaaafifa@127.0.0.1:3000/ayeshaaafifa/Assignment01.git (push)
ayeshaaafifa@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://ayeshaaafifa@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/ayeshaaafifa/Assignment01.git/'
ayeshaaafifa@ubuntu:~/Assignment01$ git remote -v
gitea   http://ayeshaaafifa@127.0.0.1:3000/ayeshaaafifa/Assignment01.git (fetch)
gitea   http://ayeshaaafifa@127.0.0.1:3000/ayeshaaafifa/Assignment01.git (push)
ayeshaaafifa@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://ayeshaaafifa@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/ayeshaaafifa/Assignment01.git/'
ayeshaaafifa@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://ayeshaaafifa@127.0.0.1:3000':
remote: Failed to authenticate user
fatal: Authentication failed for 'http://127.0.0.1:3000/ayeshaaafifa/Assignment01.git/'
ayeshaaafifa@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://ayeshaaafifa@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 250 bytes | 83.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/ayeshaaafifa/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
ayeshaaafifa@ubuntu:~/Assignment01$
```

*task1_gitea_push.png: gitea remote and successful push to main*

## Step 1.3: Verify the Gitea Repository




*task1_gitea_repository.png: public repository ayeshaaafifa/Assignment01 with rendered README*

# Task 2: Push the Same Repository to GitHub

## Step 2.1: Create the GitHub Repository and Verify Both Remotes

Created an empty public GitHub repository named Assignment01 and added it as the github remote.

```
git remote add github https://github.com/ayeshaaafifa/Assignment01.git git remote -v
```

```
1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm


ayeshaaafifa@ubuntu:~$ cd ~/Assignment01
ayeshaaafifa@ubuntu:~/Assignment01$ git remote add github https://github.com/ayeshaaafifa/Assignment01.git
ayeshaaafifa@ubuntu:~/Assignment01$ git remote -v
gitea   http://ayeshaaafifa@127.0.0.1:3000/ayeshaaafifa/Assignment01.git (fetch)
gitea   http://ayeshaaafifa@127.0.0.1:3000/ayeshaaafifa/Assignment01.git (push)
github  https://github.com/ayeshaaafifa/Assignment01.git (fetch)
github  https://github.com/ayeshaaafifa/Assignment01.git (push)
ayeshaaafifa@ubuntu:~/Assignment01$ _
```

*task2_remotes.png: gitea and github remotes, no token or password in the URLs*

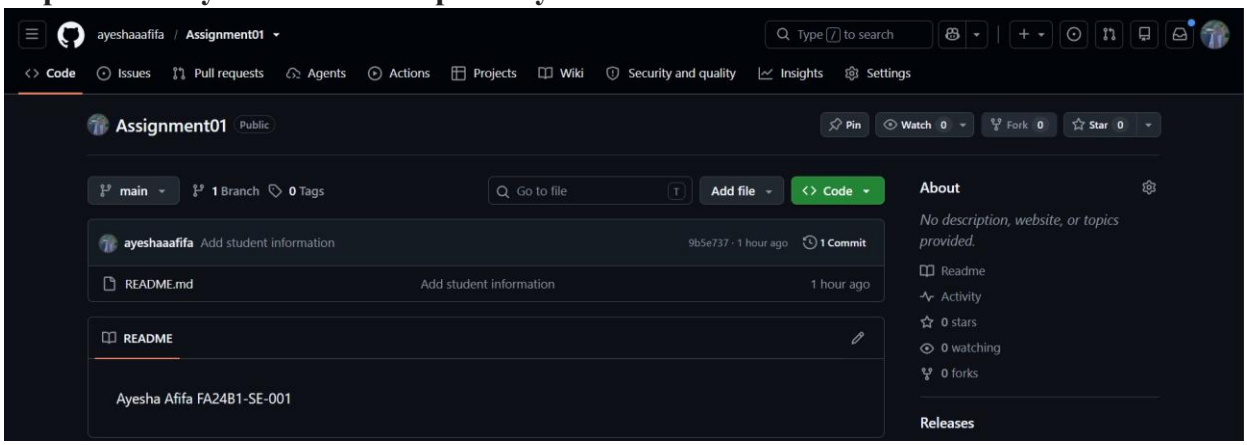
## Step 2.2: Push to GitHub

```
git push -u github main
```

```
ayeshaaafifa@ubuntu:~/Assignment01$ git remote -v
gitea   http://ayeshaaafifa@127.0.0.1:3000/ayeshaaafifa/Assignment01.git (fetch)
gitea   http://ayeshaaafifa@127.0.0.1:3000/ayeshaaafifa/Assignment01.git (push)
github  https://github.com/ayeshaaafifa/Assignment01.git (fetch)
github  https://github.com/ayeshaaafifa/Assignment01.git (push)
ayeshaaafifa@ubuntu:~/Assignment01$ git push -u github main
Username for 'https://github.com': ayeshaaafifa
Password for 'https://ayeshaaafifa@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 250 bytes | 83.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ayeshaaafifa/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'github/main'.
ayeshaaafifa@ubuntu:~/Assignment01$ _
```

*task2_github_push.png: successful push of main to GitHub*

## Step 2.3: Verify the GitHub Repository




*task2_github_repository.png: public repository ayeshaaafifa/Assignment01 with rendered README*

# Task 3: Track Three Large Files with Git LFS

## Step 3.1: Configure Git LFS

```
git lfs install git lfs version git
lfs track "*.bin" cat
.gitattributes
```

```
ayeshaaafifa@ubuntu:~/Assignment01$ cd ~/Assignment01
ayeshaaafifa@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
ayeshaaafifa@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
ayeshaaafifa@ubuntu:~/Assignment01$ git lfs track "*.bin"
"*.bin" already supported
ayeshaaafifa@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
ayeshaaafifa@ubuntu:~/Assignment01$ _
```

*task3_lfs_setup.png: Git LFS initialized, version shown, *.bin rule in .gitattributes*

## Step 3.2: Create and Verify Three LFS Files

Created three files of 101 MiB, 102 MiB and 103 MiB, staged them, and listed them with Git LFS (shown as 106, 107 and 108 MB).

```
truncate -s 101M large-file-1.bin truncate -s 102M large-file-
2.bin truncate -s 103M large-file-3.bin
git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
git lfs ls-files --size
```

```
ayeshaaafifa@ubuntu:~/Assignment01$ cd ~/Assignment01
ayeshaaafifa@ubuntu:~/Assignment01$ git lfs install
Updated Git hooks.
Git LFS initialized.
ayeshaaafifa@ubuntu:~/Assignment01$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
ayeshaaafifa@ubuntu:~/Assignment01$ git lfs track "*.bin"
"*.bin" already supported
ayeshaaafifa@ubuntu:~/Assignment01$ cat .gitattributes
*.bin filter=lfs diff=lfs merge=lfs -text
ayeshaaafifa@ubuntu:~/Assignment01$ truncate -s 101M large-file-1.bin
ayeshaaafifa@ubuntu:~/Assignment01$ truncate -s 102M large-file-2.bin
ayeshaaafifa@ubuntu:~/Assignment01$ truncate -s 103M large-file-3.bin
ayeshaaafifa@ubuntu:~/Assignment01$ git add .gitattributes large-file-1.bin large-file-2.bin large-file-3.bin
ayeshaaafifa@ubuntu:~/Assignment01$ git lfs ls-files --size
f1753a034b * large-file-1.bin (106 MB)
2338a970c0 * large-file-2.bin (107 MB)
cb91d5908e * large-file-3.bin (108 MB)
ayeshaaafifa@ubuntu:~/Assignment01$
```

*task3_lfs_files.png: three files tracked by Git LFS, each larger than 100 MiB*

## Step 3.3: Commit and Push the LFS Files

```
git commit -m "Add three files using Git LFS" git push github main
```

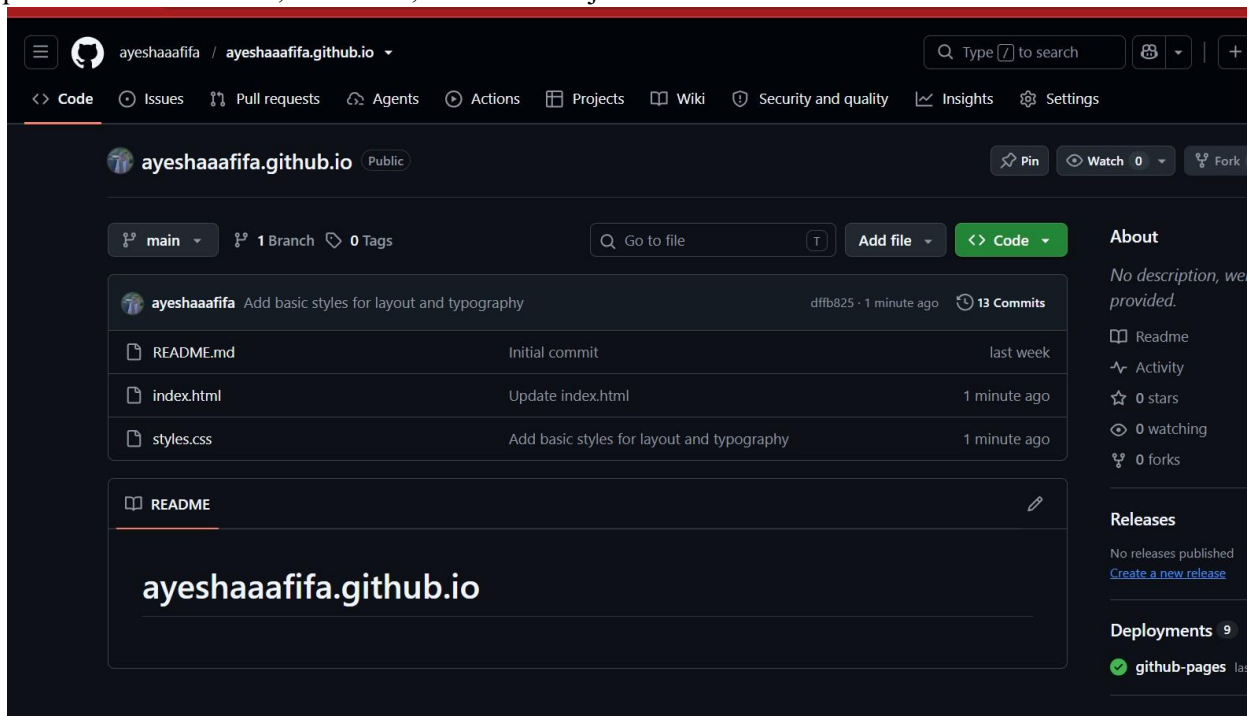
```
ayeshaaafifa@ubuntu:~/Assignment01$ cd ~/Assignment01
 push github mainayeshaaafifa@ubuntu:~/Assignment01$
ayeshaaafifa@ubuntu:~/Assignment01$ git push github main
Username for 'https://github.com': ayeshaaafifa
Password for 'https://ayeshaaafifa@github.com':
Username for 'https://github.com': ayeshaaafifa
Password for 'https://ayeshaaafifa@github.com':
Uploading LFS objects: 100% (3/3), 324 MB | 1.8 MB/s, done.
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 796 bytes | 265.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/ayeshaaafifa/Assignment01.git
   9b5e737..9816c64  main -> main
ayeshaaafifa@ubuntu:~/Assignment01$
```

*task3_lfs_push.png: LFS upload 100% (3/3) and successful push to GitHub*

# Task 4: Create a Portfolio or CV with GitHub Pages

## Step 4.1: Create and Push the Portfolio Repository

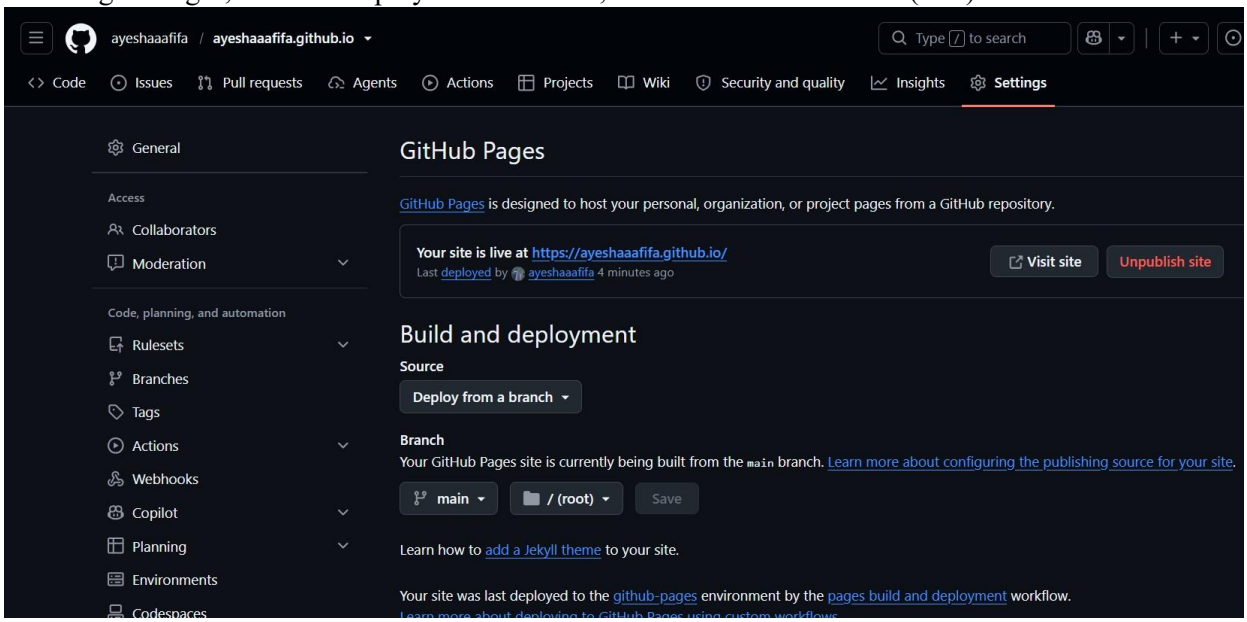
Used the public repository ayeshaaafifa.github.io with index.html and styles.css on the main branch. The portfolio has About Me, Education, Skills and Projects sections.




*task4_pages_repository.png: repository with index.html and styles.css*

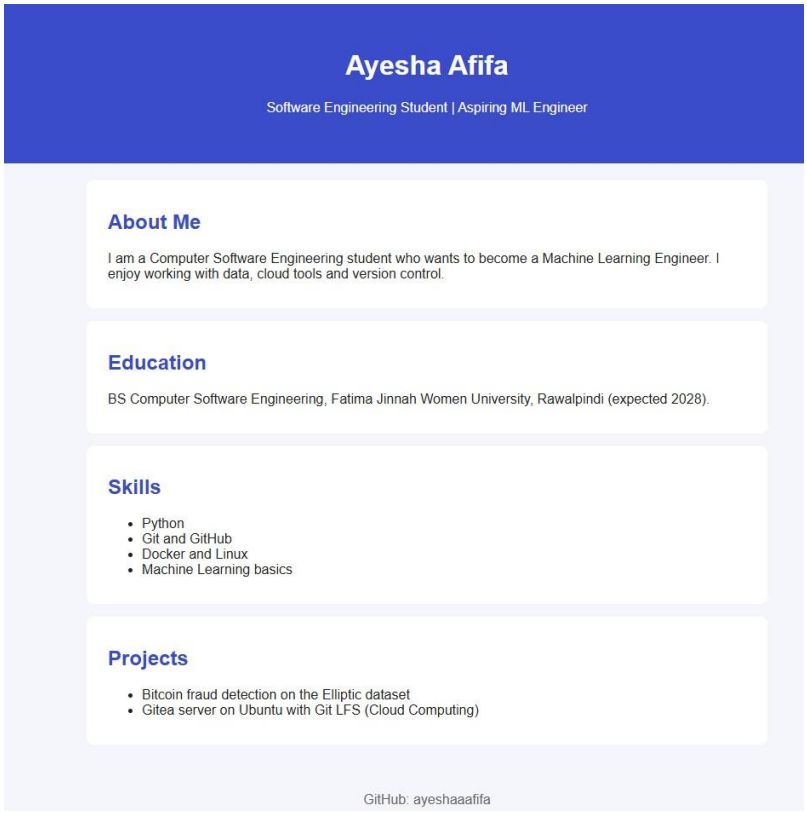
## Step 4.2: Publish with GitHub Pages

In Settings > Pages, selected Deploy from a branch, branch main and folder / (root).




*task4_pages_deployment.png: GitHub Pages live at https://ayeshaaafifa.github.io/*

## Step 4.3: Verify the Live Portfolio




*task4_portfolio_live.png: live portfolio at https://ayeshaaafifa.github.io/*